Both 62-bit targets recovered on one GPU

Frozen protocol-v11 completed all 12 prescribed executions on one RTX PRO 6000 Blackwell Server Edition. Every result passed fresh CPU certificate replay and scalar-corruption rejection. The largest complete prescribed rung is 62 field bits. This is a bounded implementation result, not an absolute hardware maximum, certified speedup, matched-rho result or security conclusion. The relation oracle is generic meet-in-the-middle; it is not the paper's conditional tensor backend.

Matched 52-bit controls

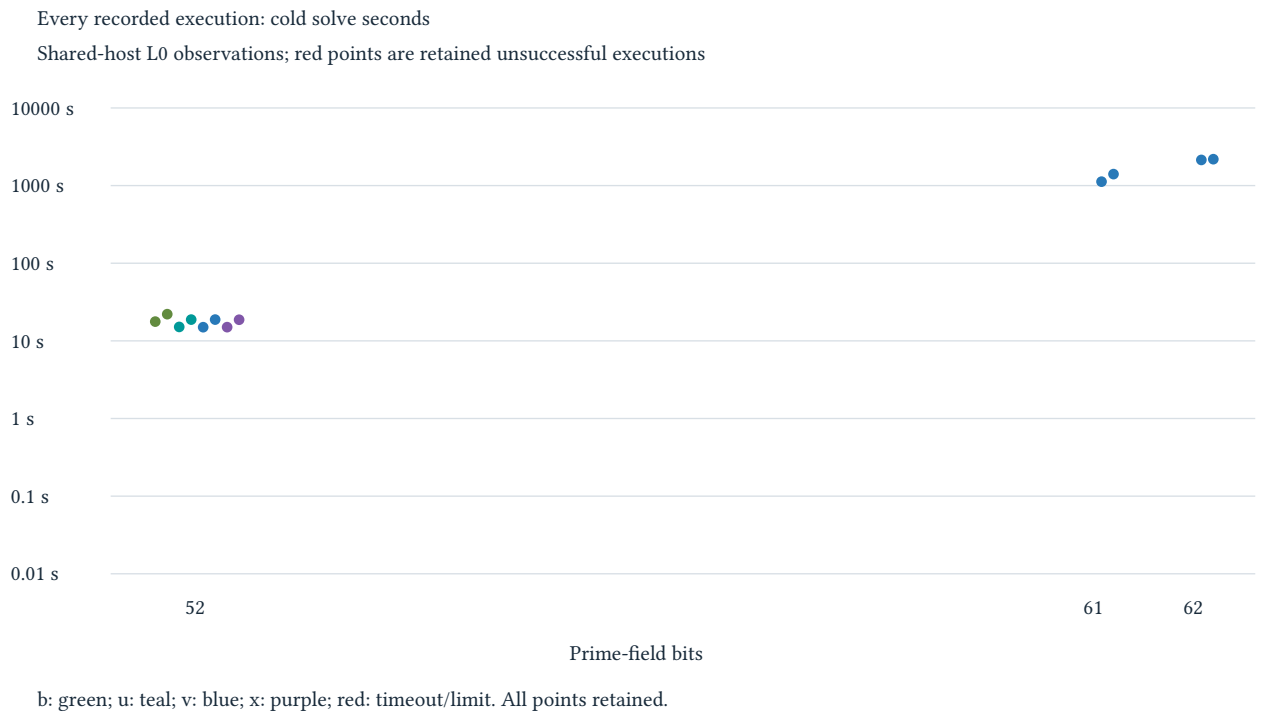
Bits	Seed	b: 16-byte Thrust	u: 12-byte Thrust	v: 12-byte CUB	x: identical v
52	101	17.700446	15.116574	14.998561	15.019326
52	202	22.070081	18.789266	18.797276	18.710886

Cold solve seconds, rounded from `results-v11/rows.csv`. Arms run b/u/v/x for seed 101 and x/v/u/b for seed 202. Certificates match byte for byte across arms. All clocks are shared-host L0 diagnostics, with wall_valid=false; no inferential interval or certified timing ratio is available.

Larger fixed targets: dense CUB arm v

Bits	Seed	Cold solve (s)	With audit (s)	Relations	Attempts
61	101	1123.435000	1123.454534	1617	2292
61	202	1405.348145	1405.366104	1591	2432
62	101	2139.880952	2139.908483	1724	2675
62	202	2187.942595	2187.971582	1771	2485

At 62 bits, oracle time was 2137.956289 / 2185.870989 seconds and linear elimination 1.618892 / 1.757169 seconds. Table admission estimates were 48,798,250,775 / 49,539,778,479 bytes, including queried sort scratch; they are not measured peak VRAM. No prescribed cell timed out, refused admission or failed.



Every recorded execution is shown. Colors denote arms, not independent hardware.

Contract, validation and custody

Measured source: `d44b59149f468357e86186f956ecb7e323f38915`. Pod: `jees49ew9dkfd7`, created 2026-10-30T15:50:53 UTC. NVIDIA telemetry reports 97,887 MiB, driver 595.91.07 and compute capability 12.0. The build used CUDA 13.2, sm_120 and Rust 1.88.0. Provider rate was USD 2.49/hour; this is not a final bill. The controller verified archive integrity and pod deletion after completion. A subsequent complete provider listing independently confirmed absence.

Command: `bash research/explicit_relations_e2e_20261030/run-sort.sh`, with EPHEMERAL_OUT=/results. Seeds 101/202; m=8; tau=2ceil(r^(1/8))+2; `rows <= 2^32`, `work <= 2^34`, chunk 2^20, columns 16384, trials 16. Cold solve caps: 180 seconds at 52 bits, 2700 at 61, 5400 at 62. Fresh replay: 60 seconds; PARI fixture generation: 180; worker: 18000; controller: 18300. The plan finishes both targets at a size, then stops larger sizes if either is incomplete. Its complete maximum is 12 executions, all recorded here.

The source accepts certified prime-order curves with `p < 2^62` and `r < 2^63`. Public targets are coordinate lifts; no target scalar is generated or supplied. The driver checks the curve, primality and the unique Hasse-interval order witness. All 19 Rust tests, host/device differential checks, Clippy, maximum-cap protocol headers and recorded query/service sanitizer checks passed. Every successful certificate passed fresh CPU replay and rejected scalar+1. There is no root crypto plug-in build on the constrained local workspace.

Cold solve includes process startup, validation, all base construction and relation attempts, elimination, extraction, in-process replay, serialization and teardown. The separate audit total also includes target generation, fresh replay and negative control. Build/preflight and curve fixture generation are outside cold solve. CPU user/system time and measured peak VRAM were not captured.

Frozen protocol hash: `0f916e48bea43b0b7dabb85bc8cb50fe0d31c2a268aa89c016e1df6b46a0a09e` Raw archive SHA256: `e80cf3fb2c64fd2510e1eac7f15d0477b3a3df9408dfd88d73df0b6e3a888372` Archive: `s3://setec-astronomy-artifacts/validation/ecbench-explicit-x/e2e-20261030-v11/results.tgz`. Raw records were committed and pushed in `099dff2edd81` before this report. The archive manifest hashes every member; certificate-audit.json retains native replay and rejection outputs. These manifests do not replace replay.

Schema, flow and file hashes pass. Native analysis and the strict coverage audit pass; the latter rejects eight deliberate coverage/status/input mutations. The canonical checker retains five rectangular-coverage findings: 24 assumed rows versus 12, b/u/x each cover one rather than three sizes, and arm instance sets differ. The frozen plan prescribes four arms only at 52 bits and v alone at larger sizes. This is not a full canonical-checker pass or independent scientific validation. See `results-v11/CHECKER-NOTE.md`; no row was excluded.

Implementation context and next steps

Exact 12-byte rows and queried CUB scratch are described in DENSE-ROWS.md, which cites NVIDIA alignment guidance and the CCCL DeviceMergeSort source. V10 was never executed. The separately frozen v13 adds persistent queries and full right-sum caching with distinct controls; its results are not included here. A separate 63-bit arithmetic extension remains pending GPU validation. Neither that source change nor this complete 62-bit panel establishes a hardware maximum.